L-C-566c

SEPTEMBER 17, 1957

SUPERSEDING
Int. Fed. Spec. L–C–00566b (GSA–FSS)
August 15, 1956, and
Fed. Spec. L–C–566a
April 15, 1952

FEDERAL SPECIFICATION

COMB, HAIR (PLASTIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers hair combs made of thermoplastic material.

1.2 Classification.

1.2.1 *Type.*—There shall be but one type of combs as specified hereinafter.

1.2.2 *Grade.*—Combs shall be of first grade only.

1.2.3 *Sizes.*—Combs shall be of the following sizes:

Size A.—Long (7 – 7½ inches)
Size B.—Short (4½ – 5 inches)

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

QQ–S–781—Strapping, Flat, Steel.
PPP–B–566 — Boxes, Folding, Paperboard.
PPP–B–636—Boxes, Fiber.
PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Miltary Specifications:

MIL–L–10547 — Liners, Case, Waterproof.

441882/57/No 78

FSC 8530

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military specifications and standards, required by contractors in connection with specific procurement functions, should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—Material shall be of virgin thermoplastic molding material, shall be clean, smooth, and free from buffing compound, flash, or foreign matter.

3.2 Design.—Combs shall be coarse and fine toothed over approximately equal lengths. The back shall be oval or tapered, as specified in the invitation for bids. The "gate" shall be adequate and located in the coarse-toothed section of the comb.

3.3 Dimensions.

3.3.1 *Size A, long.*—Combs shall be 7 to 7½ inches long. The width at the widest point shall be not less than 1⅛ inches and at the narrowest point not less than ⅞ inch. The thickness of back at thickest point shall be not less than $^{13}/_{64}$ inch. The coarse section shall contain from 9 to 10 teeth per inch and be in the wide section of the comb. The fine section shall contain from 17 to 19 teeth per inch and shall be in the narrow section of the comb.

3.3.2 *Size B, short.*—Combs shall be 4½ to 5 inches long. The width at the widest point shall be not less than 1⅛ inches and at the narrowest point not less than ⅞ inch. The thickness of back shall be $^{8}/_{64}$ to $^{11}/_{64}$ inch. The coarse section shall contain from 8½ to 10 teeth per inch and be in the wide section of the comb. The fine section shall contain from 16 to 19 teeth per inch and be in the narrow section of the comb.

3.4 Color. — Unless otherwise specified, combs shall be black and highly polished.

3.5 Deformation in hot water.—The comb shall maintain its shape when immersed for 30 minutes in water kept at 160° ± 2°F. The teeth shall not bend nor contract visibly when subjected to this treatment (see 4.4.2).

3.6 Stiffness of teeth.—The coarse teeth shall be capable of lifting a 1½ pound weight and the fine teeth shall be capable of lifting a 1 pound weight under the conditions prescribed in 4.4.3.

3.7 Thumb test.—The teeth of the comb shall not break out nor be permanently deformed when subjected to the "thumb" test described in 4.4.4.

3.8 Flammability. — The comb shall not burn at a rate greater than 2.0 inches per minute when tested as described in 4.4.5.

3.9 Accelerated aging test. — The comb shall not undergo appreciable changes in physical characteristics and shall meet the "thumb" test requirement specified in 3.7 after it is subjected to the accelerated aging treatment described in 4.4.6.

3.10 Workmanship.—Combs shall be clean and well made, and shall be free from any defects which affect appearance and quality, or which may affect serviceability. There shall be no chipping or visible fracture of material where the "gate" is cut. The combs shall be free of warpage. The teeth shall be uniform in length, shall have an adequate fillet on the sides and at the roots, shall be straight and truly aligned, and shall be not more than 0.002 inch out of register. The teeth shall not have pronounced, sharp, or rough miter lines, and the points or ends

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shall be rounded and have no sharp projecting fins. After finishing, the comb shall be clean, smooth, and shall be free from buffing compound, flash or foreign matter.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of all combs offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be taken from each lot in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptable Quality Level (A.Q.L.) = 2.5 percent defective to verify compliance with all stipulations of this specification regarding closure, marking, and other requirements not involving tests.

4.1.2.1 *Military Standard MIL–STD–105.*—When Military Standard MIL–STD–105 specifies an action by the Government, it shall at the option of the Government be performed either by the Government or by the contractor under the supervision of the Government inspector.

4.1.3 *Sampling for tests.*—From each inspection lot a single package containing 12 combs shall be selected at random which shall be sealed, marked and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.2 Inspection.

4.2.1 *Inspection of filled containers.*—Each sample filled container selected in accordance with 4.1.2 shall be examined for defects of the container and the closure and for unsatisfactory markings. Any container in the sample having one or more defects shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105 the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided that the contractor has removed or repaired all nonconforming containers.

4.3 **Lot acceptance tests.** — The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.4. If the specimens fail in one or more of these tests the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

4.4 Test procedures.

4.4.1 *Conditioning of test specimens.*—Unless otherwise specified, the sample combs used in the following test methods shall be conditioned for 48 hours at 23°C. ± 1.1°C. (73.5° ± 2°F.) and 50 ± 4 percent relative humidity.

4.4.2 *Deformation in hot water.*—One comb from each sample package shall be immersed for 30 minutes in water kept at 160° ± 2°F. Each comb shall be supported by a cord or wire attached to the coarse-tooth end of the comb. After the 30-minute immersion, the combs shall be examined for warping, bent teeth, or contraction due to unmolding.

4.4.3 *Stiffness of teeth.*—To one comb from each package, a 1½-pound weight shall be attached ½ inch from the root of one of the coarse teeth by means of a looped wire. The tooth shall support this load when the weight is lifted by raising the comb. Under similar conditions a fine tooth shall support a load of 1 pound.

4.4.4 *Thumb test.*—The back of one comb from each sample package shall be held in the palm of the hand and pressure applied against the fine teeth by the thumb in the

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direction of the index finger. A similar test shall be applied to the comb at minus 15°F. after the comb has been cooled to that temperature. This test shall be made on the sample combs taken from deliveries in comparison with the bid samples.

4.4.5 *Flammability.*—The back of the comb shall be marked with a line opposite the first coarse tooth next to the fine-tooth section and with another line 3 inches distant from the first in the direction of the fine-tooth section. The coarse tooth section of the comb shall be clamped in a support with the longitudinal axis of the comb horizontal and its transverse axis inclined at 45 degrees to the horizontal. Under the comb there shall be clamped a piece of 20-mesh Bunsen-burner gauze in a horizontal position about ¼ inch below the ends of the teeth of the comb and with about ½ inch of the fine-tooth section of the comb extending beyond the edge of the gauze. A Bunsen burner or an alcohol lamp with a flame ½ to ¾ inch in height shall be placed under the free end of the comb and adjusted so that the flame tip is just in contact with the specimen. At the end of 30 seconds the flame shall be removed. A stop watch shall be started when the flame reaches the first mark on the back of the comb and stopped when the flame reaches the mark on the back of the comb 3 inches distant from the first. In case the comb does not continue to burn after the first ignition, the burner shall be replaced under it for a second period of 30 seconds immediately following the extinction of the flame. If the comb does not continue burning after the second ignition, it shall be reported as self-extinguishing.

4.4.6 *Accelerated aging test.*—One comb from each sample package shall be heated for 48 hours at a temperature of 160°F. in a circulating-air oven. After cooling to room temperature the comb shall be examined for changes in physical characteristics, and shall be subjected to the "thumb" test described in 4.4.4.

5. PREPARATION FOR DELIVERY

The definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*

5.1.1.1 Twelve combs of one size only shall each be wrapped or placed in a bag of suitable thin paper so the combs are separated to prevent scratching of any surfaces, and packaged either in a snug-fitting two-piece full telescope set-up box made of nonbending paperboard having a thickness of not less than 0.036 inch and made in compliance with Federal Specification PPP–B–676 type I, class A; or in a folding carton of bending paperboard having a thickness of not less than 0.022 inch made in compliance with Federal Specification PPP–B–566 style II, type A, class A, with locked inner flaps, or style II with one or both ends of the box having twin slot lock ends. (See fig. 2 of Federal Specification PPP–B–566).

5.1.1.2 *Intermediate package.*—Thirty-six packages, each containing twelve combs, shall be packaged in a corrugated or solid fiberboard container constructed in accordance with Federal Specification PPP–B–636, type I, class 1, style RSC. All flaps shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps.

5.1.2 *Level C.*—Combs shall be packaged in accordance with commercial practice.

5.2 Packing.

5.2.1 *Level A.*—One thousand two hundred ninety-six combs shall be packed in a shipping container conforming to Federal Specification PPP–B–636, type I or II, class 2, style RSC, compliance symbol V3s or V3c.

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Each shipping container shall be provided with a case liner conforming to Military Specification MIL–L–10547, type I or II, grade B or C, class 2. The boxes shall be closed and strapped in conformance with the Appendix to the box specification, except the straps shall be type I, class B, grade 2, flat steel of Federal Specification QQ–S–781.

5.2.2 *Level B.*—One thousand two hundred ninety-six combs shall be packed in a shipping container constructed to conform to Federal Specification PPP–B–636, type I or II, class 1, style RSC. All flaps of the fiberboard containers shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 *Level C.*—Unless otherwise specified, the product shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Civil agencies.*—Shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* — Interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) The size, design and color of combs, if other than black, is desired. (See 1.2.3, 3.2 and 3.4).

(*c*) Levels of packaging and packing required (see 5).

6.2 Transportation description.—Transportation description applicable to these items is:

Combs, NOIBN, other than silver trimmed in boxes.

Carload weight 20,000 lbs., subject to Rule 34, or C.F.C.

Truckload weight 20,000 lbs.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q
Navy—MC
Air Force.

Copies of this specification may be purchased for 5 cents each.

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